Light

GCSE Computer Science — Programming Data Structures (Set 3)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

A variable declared outside all functions is called a:

- ○ **A.** Local variable
- ○ **B.** Global variable
- ○ **C.** Constant variable
- ○ **D.** Private variable

Q2 (1 mark)

What is a key benefit of using a named constant like `PI = 3.14159` instead of typing 3.14159 throughout a program?

- ○ **A.** If the value needs updating, you only change it in one place, and the name makes the code more readable
- ○ **B.** It makes the program run faster
- ○ **C.** It uses less memory than a variable
- ○ **D.** It allows the value to be changed while the program runs

Q3 (1 mark)

A program needs to store the year a student was born (e.g., 2008). Which data type is most appropriate?

- ○ **A.** Integer
- ○ **B.** Real
- ○ **C.** String
- ○ **D.** Boolean

Q4 (1 mark)

In a game, a variable `isGameOver` tracks whether the game has ended. Which data type should this variable use?

- ○ **A.** String
- ○ **B.** Integer
- ○ **C.** Real
- ○ **D.** Boolean

Q5 (1 mark)

A fitness app records a user's height as 1.75 metres. Which data type stores this correctly?

- ○ **A.** Boolean
- ○ **B.** Character
- ○ **C.** Real
- ○ **D.** Integer

Q6 (1 mark)

An array has 8 elements. What is the index of the last element?

- ○ **A.** 8
- ○ **B.** 9
- ○ **C.** 1
- ○ **D.** 7

Q7 (1 mark)

An array is declared as: `fruits = ['Apple', 'Banana', 'Cherry', 'Date']`
What value does `fruits.length` return?

- ○ **A.** 3
- ○ **B.** 5

- ○ **C.** 4
- ○ **D.** 0

Q8 (1 mark)

Given `phone = '07700123456'`, what is the output of `phone.right(6)`?

- ○ **A.** 077001
- ○ **B.** 07700
- ○ **C.** 456
- ○ **D.** 123456

Q9 (1 mark)

What does the `float()` function do?

- ○ **A.** Converts a value to a decimal (real) number
- ○ **B.** Rounds a number to the nearest integer
- ○ **C.** Checks if a number can float on water
- ○ **D.** Removes all decimal places from a number

Q10 (1 mark)

What is the length of an empty string `''`?

- **A.** 1
- **B.** Undefined
- **C.** -1
- **D.** 0

Q11 (1 mark)

A program stores a UK National Insurance number like 'AB123456C'. Which data type is most appropriate?

- **A.** Integer
- **B.** Real
- **C.** Boolean
- **D.** String

Q12 (1 mark)

How does a record differ from an array?

- **A.** Records can only store numbers, arrays can store any type
- **B.** Records store a fixed number of items, arrays can grow indefinitely
- **C.** A record groups different data types with named fields; an array stores multiple items of the same type accessed by index
- **D.** There is no difference; the terms are interchangeable

Q13 (2 marks)

Which statement correctly creates an array containing the numbers 5, 10, 15, and 20?

- A. `numbers = (5, 10, 15, 20)`
- B. `numbers = [5, 10, 15, 20]`
- C. `numbers = {5, 10, 15, 20}`
- D. `numbers = 5, 10, 15, 20`

Q14 (2 marks)

Given `message = 'hello world'`, what is the output of `message.upper.left(5)`?

- A. HELLO
- B. hello
- C. HELLO WORLD
- D. world

Q15 (2 marks)

What is the result of `int(7.9)`?

- A. 8
- B. 7.9
- C. Error

- D. 7

Q16 (2 marks)

A 2D array is declared as `table[4][6]`. How many elements does this array contain in total?

- A. 10
- B. 24
- C. 46
- D. 2

Q17 (2 marks)

In most programming languages, what is the result of comparing `'Apple' == 'apple'`?

- A. True, because they contain the same letters
- B. Error, because strings cannot be compared
- C. True, because both are fruit names
- D. False, because string comparison is case-sensitive

Q18 (2 marks)

An algorithm needs to count how many times the value 5 appears in an array. Which approach is correct?

- A. Initialise a counter to 0, loop through each element, increment

the counter whenever an element equals 5

- ○ **B.** Use array.length to count occurrences of 5
- ○ **C.** Check only the first and last elements for the value 5
- ○ **D.** Sort the array first, then the count will be automatically calculated

Q19 (2 marks)

In many programming languages, what does `bool(0)` return?

- ○ **A.** 0
- ○ **B.** True
- ○ **C.** False
- ○ **D.** Error

Q20 (2 marks)

A string operation returns the position of the first occurrence of a character within a string. For `'programming'.find('g')`, what position would be returned?

- ○ **A.** 0
- ○ **B.** 3
- ○ **C.** 4
- ○ **D.** 10

Q21 (2 marks)

Given a 2D array:
`matrix = [[1, 2], [3, 4], [5, 6]]`
What statement changes the value 4 to 10?

- A. `matrix[4] = 10`
- B. `matrix[1][1] = 10`
- C. `matrix[2][2] = 10`
- D. `matrix[1, 1] = 10`

Q22 (2 marks)

A record `car` has fields: make, model, year, price. Which expression correctly displays the make and year?

- A. `print(car[make], car[year])`
- B. `print(car.make, car.year)`
- C. `print(make.car, year.car)`
- D. `print(car(make, year))`

Q23 (2 marks)

Inside a function, a programmer wants to modify a global variable `score`. In many languages, what keyword indicates this intention?

- A. local
- B. private

- **C.** global
- **D.** public

Q24 (3 marks)

In bubble sort, one complete pass through the array [5, 3, 8, 1] compares adjacent elements and swaps them if they are in the wrong order. Analyse what the array looks like after exactly one complete pass (comparing elements left to right).

- **A.** [1, 3, 5, 8]
- **B.** [3, 5, 1, 8]
- **C.** [3, 5, 8, 1]
- **D.** [5, 3, 1, 8]

Q25 (3 marks)

A programmer needs to reverse a string so 'HELLO' becomes 'OLLEH'. Analyse which algorithm correctly achieves this.

- **A.** Use .upper to reverse the string
- **B.** Start with an empty result string; loop through the original string from the last character to the first, concatenating each character to the result
- **C.** Swap the first and last characters only
- **D.** Use .length to reverse the characters

Q26 (3 marks)

A 2D array stores sales data: 3 rows (products) and 4 columns (quarters).

```
sales = [[100, 150, 200, 180],
         [90, 120, 140, 160],
         [80, 110, 130, 150]]
```

Analyse how to calculate the total sales for Quarter 2 (column index 1).

- ○ **A.** Add sales[1][0] + sales[1][1] + sales[1][2] + sales[1][3]
- ○ **B.** Use sales[1].length to get the Q2 total
- ○ **C.** Loop through each row (0 to 2) and add sales[row][1] to a running total, giving 150 + 120 + 110 = 380
- ○ **D.** Add all elements in the array together

Q27 (3 marks)

An array of employee records stores: name (string), department (string), salary (integer).
Analyse how to find all employees in the 'Sales' department earning over £30000.

- ○ **A.** Check only the first employee record in the array
- ○ **B.** Loop through all records; for each record, check if department equals 'Sales' AND salary > 30000; when both conditions are true, include that employee in the results
- ○ **C.** Sort the array by salary then check the first few records
- ○ **D.** Use employees.department to get all Sales employees

Q28 (3 marks)

A registration form collects: full name, email address, age, and whether the user wants to receive newsletters.

Analyse the most appropriate data type for each field.

- ○ **A.** All fields should use string type for consistency
- ○ **B.** Name: string, Email: integer, Age: real, Newsletter: character
- ○ **C.** Use arrays to store each field
- ○ **D.** Name: string, Email: string, Age: integer, Newsletter: Boolean - each type matches the nature of the data it stores

Q29 (3 marks)

An array contains [10, 20, 30, 40, 50]. A program needs to remove the element at index 2 (value 30), shifting remaining elements left.
Analyse the correct result and approach.

- ○ **A.** The result is [10, 20, 40, 50] after shifting elements at indices 3 and 4 one position left to fill the gap, then reducing the array length
- ○ **B.** The result is [10, 20, 0, 40, 50] with a zero replacing the removed element
- ○ **C.** The result is [10, 20, 30, 40] by removing the last element instead
- ○ **D.** Elements cannot be removed from arrays

Q30 (3 marks)

A program needs to count the number of words in a sentence. Words are separated by spaces.
For the sentence 'The quick brown fox', analyse an algorithm to count the words.

- ○ **A.** Count the total number of characters and divide by 5

○ **B.** Use .length to get the word count directly

○ **C.** Count the number of space characters and add 1 - the sentence has 3 spaces, so 3 + 1 = 4 words; alternatively, loop through splitting at spaces and count the resulting segments

○ **D.** Count capital letters to determine word count

Total: **55** marks • Time: **30 min**

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